UNIT 1: ATOMIC STRUCTURE

H. Hydrogen (18.5)
I. Alkali Metals (18.1, 18.2)

UNIT 2: BONDING

A. Ionic, Covalent, Metallic Bonding (7.1, 7.5, 7.2)
B. Covalent Bonding
    1. Bond Energy (7.5)
    2. Electronegativity (7.2)
C. Lewis Structures
    1. Lewis structures and Formal Charge (7.3)

- Quiz 2 OPEN - DUE ON OCT. 1
- Lesson on Atomic Properties & Chemical Reactivity
- STUDY SHEETS - Group 1 & 2 metals/halogens
  - Reactivity of Alkali Metals

LEARNING OUTCOMES

Utilize different CHEM 103 learning outcomes to predict reactivity and physical properties of main group elements and compounds.

Differentiate between different types of bonding, including covalent, ionic, and metallic bonding.

Generate Lewis structures for covalent molecules and ions and, coupled with VSEPR and Valence Bond Theory, determine the molecule's geometry, polarity, and bonding (bond order, bond length, strength and energy).

The more common practice currently is to produce H_2 by *thermal methods*

$$CH_{4(g)} + H_2O_{(g)} \xrightarrow[\text{catalyst}]{\text{high T}} CO_{(g)} + 3H_{2(g)}$$

CH_4 = Natural gas

Steam-reforming process, high temperatures, Ni-based catalyst

$$CO_{(g)} + H_2O_{(g)} \xrightarrow[\text{catalyst}]{\text{high T}} CO_{2(g)} + H_{2(g)}$$

The water-gas shift reaction, with catalyst

CO further oxidizes to CO_2 & produces H_2.

See video next page for example of steam reforming to produce H_2 for Haber process.

* Least expensive way to produce H_2
* Most common way to produce H_2
* produces CO_2 (disadvantage). Some methods remove and store the CO_2 as it is formed. See video next page.

(b) Uses

(1) Haber Process: Ammonia is synthesized using N_2 and H_2.

- Ammonia ranks first, on a mole basis, among compounds produced industrially via the Haber process.

$$N_2(g) + 3H_2(g) \rightleftharpoons 2\,NH_3(g)$$

- Watch this video from the Royal Society of Chemistry on ammonia production, starting with CH_4 and utilizing the steam-reforming process and the water-gas shift reaction to produce the H_2 needed for the Haber process.

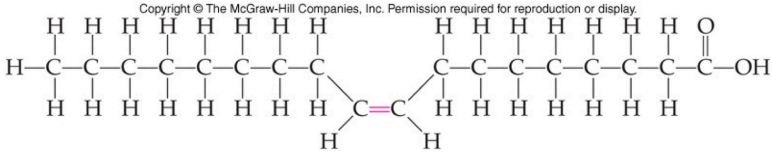
https://youtu.be/wiwl4eoHbig

(2) Hydrogenation

Food Industry: Processing of vegetable oils. Hydrogenation converts double bonds to single bonds and the properties of the oil change (melting point increases, and they become solids or semi-solids).

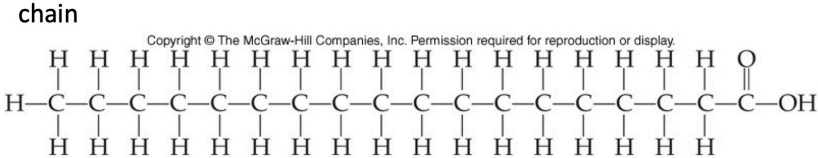
- Oleic acid: a typical unsaturated fatty acid with 18 carbons in the chain

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- Stearic acid: a typical saturated fatty acid with 18 carbons in the chain

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Petroleum, plastics, and food industries employ hydrogenation on an large scale

Denniston, K.J.; Topping, J.J.; Dorr, D.R.; Caret, R.L. General, Organic, and Biochemistry; 9th Ed.; McGraw Hill: New York; 2014.

(3) An alternative fuel to hydrocarbons?

Compare the combustion reactions of hydrogen and hydrocarbons (like octane), both of which are highly exothermic:

$H_2(g) + O_2(g) \rightarrow$ $\Delta H^o = -242$ kJ/mol

$C_8H_{18}(g) + O_2(g) \rightarrow$ $\Delta H^o = -5200$ kJ/mol

Evaluate the advantages and disadvantages of using hydrogen as a possible fuel:

advantages	disadvantages

(c) Reactivity

Given its small size and electron configuration of $1s^1$, hydrogen exhibits unique chemical behaviour.

(1) It behaves like an alkali metal in adopting a +1 oxidation state:

nonmetals + H_2 →

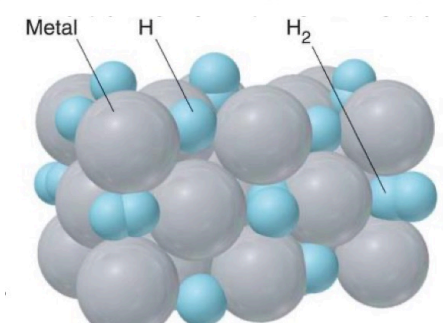
(2) It behaves like a halogen in adopting a –1 oxidation state:

group 1 or 2 metals + H_2 →

(3) It reacts with many transition metals to form nonstoichiometric **metallic hydrides (interstitial hydrides)**. H_2 molecules and H atoms occupy the holes in the crystal structure of the metal

transition metals + H_2 →
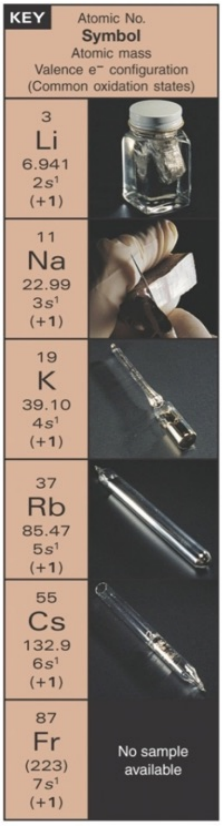
Figure 13.2 A metallic (interstitial) hydride.

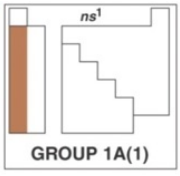


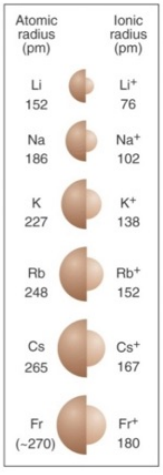
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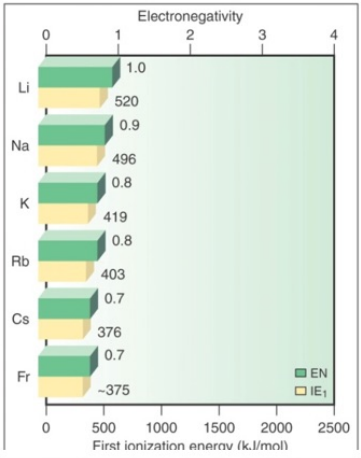
I. Alkali Metals (18.1, 18.2)

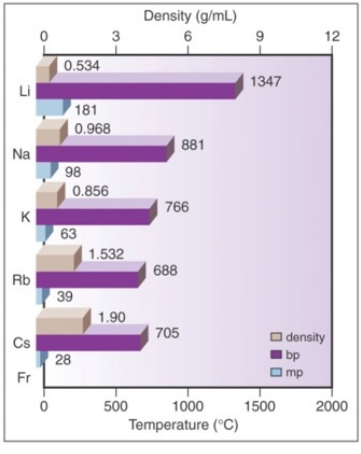
Atomic and physical properties of the alkali metals.







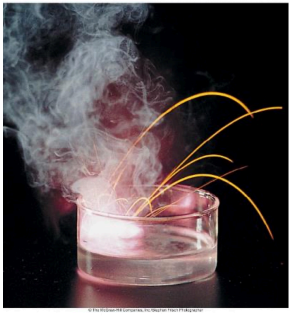




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Lithium floating in oil floating on water. Alkali metals have low densities.

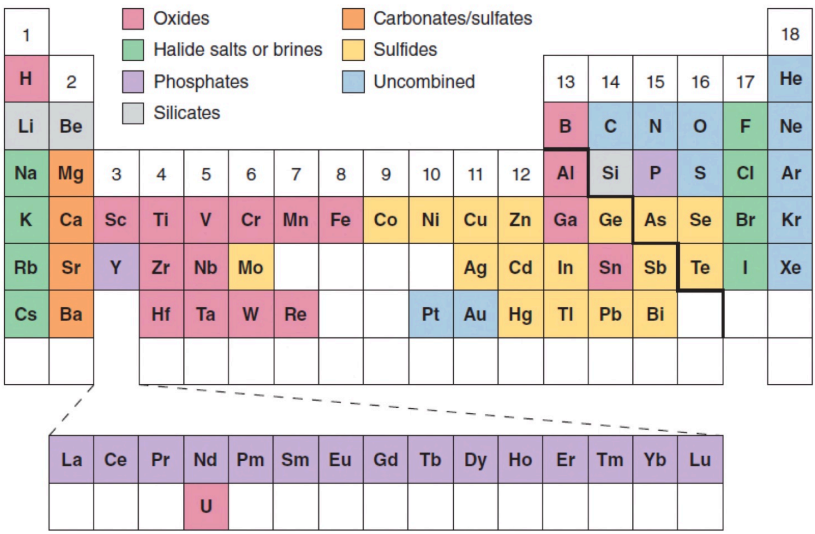


Potassium reacting with water. Alkali metals are very reactive.

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—Most elements occur in ores, natural compounds or mixtures from which an element must be extracted.

—Alkali metals always occur in nature as **+1 cations** rather than as free metals.



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Figure 23.4 **Sources of the elements.**

Uses of alkali-metal compounds:

Sources of alkali and alkaline-earth metals


spodumene


halite


sylvite


beryl


magnesite


limestone


celestine

webmineral.com

Sodium ore is called ***halite***, which is obtained either by the evaporation of brines or by mining salt deposits.

Na is **extracted** by reduction. Molten NaCl is used and the Na+ is reduced to Na. Also Cl- is oxidized to Cl_2 (g)

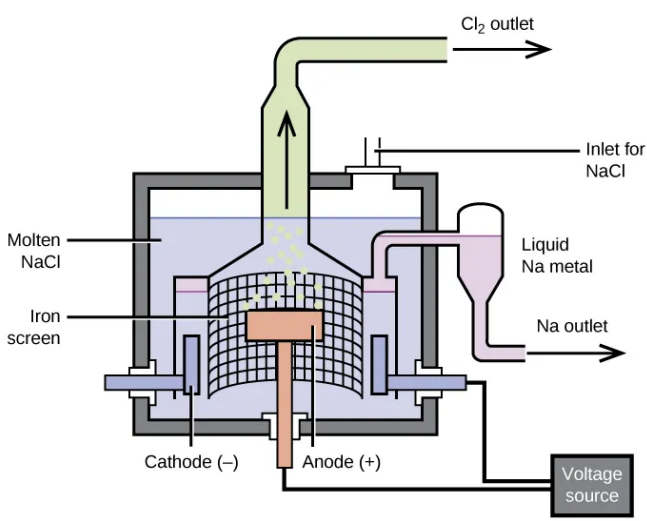


Figure 18.10 Pure sodium metal is isolated by electrolysis of molten sodium chloride using a Downs cell.

STUDY SHEET Solutions Posted **under SEPT 22_24 PreLec Notes**

Describe the reactivity of alkali metals (M) with nonmetals, water, hydrogen, oxygen.

Write **balanced** reactions for each case below. Provide states of matter (g, l, s). What happens to the oxidation state of the alkali metal in all cases?

$M (s) + X_2 \rightarrow$ X_2 = halogen

$M (s) + H_2O (l) \rightarrow$

$M (s) + H_2 (g) \rightarrow$

$M (s) + O_2 (g) \rightarrow$

Write reactions of alkali metal oxides + water. What type of oxides are the alkali metal oxides?

STUDY SHEET **Answers posted under this week's notes in our Pre-Lec Notes module**

Describe the reactivity of alkali metals (M) with nonmetals, water, hydrogen, oxygen.

Write **balanced** reactions for each case below. Provide states of matter (g, l, s). What happens to the oxidation state of the alkali metal in all cases?

halogen $2M(s) + X_2 \rightarrow 2MX(s)$

ex. NaCl, KCl, KF, etc.

X_2 = halogen

we can predict $M \rightarrow M^+$! $X_2 \rightarrow 2X^-$

water $2M(s) + 2H_2O(l) \rightarrow 2M^+_{(aq)} + 2OH^-_{(aq)} + H_{2(g)}$

or $2MOH_{(aq)} + H_{2(g)}$

H_2 $2M(s) + H_2(g) \rightarrow 2MH_{(s)}$ ionic hydride!

O_2 $M(s) + O_2(g) \rightarrow$ Products may differ depending on conditions and metal

$4Li(s) + O_{2(g)} \rightarrow 2Li_2O$ (oxide oxygen in -2 oxid. state)

$2Na(s) + O_{2(g)} \rightarrow Na_2O_2(s)$ (peroxide oxygen in -1 oxid. state)

or $K(s)$, or $Rb(s)$, or $Cs(s) + O_{2(g)} \rightarrow KO_2$, RbO_2, CsO_2 (superoxide oxygen in $-\frac{1}{2}$ oxid. state)

Write reactions of alkali metal oxides + water. What type of oxides are the alkali metal oxides? – basic oxides

General rxn for oxide (O^{2-}) $M_2O_{(s)} + H_2O_{(l)} \rightarrow 2MOH_{(aq)}$ basic

UNIT 2: BONDING

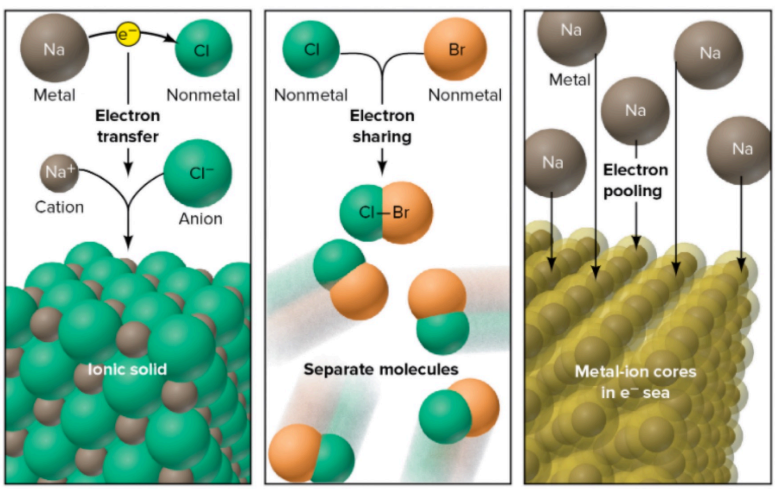
A. General (7.1, 7.2, 7.5)

Types of Bonding

Simplified view:

Metal + nonmetal	**Ionic**
Nonmetal + nonmetal	**Covalent**
Metal + metal	**Metallic bonds**

Figure 8.3 Three models of chemical bonding.



A Ionic bonding B Covalent bonding C Metallic bonding

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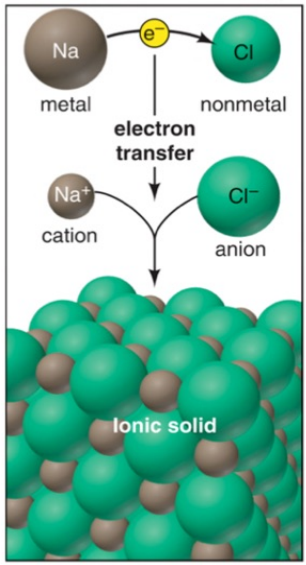
1. Ionic Bonding (Sec. 7.1, 7.5)

- metals + nonmetals
- electrons are transferred completely from one atom to another
- cations and anions, arranged to form, not molecules, but an ***extended structure***

Example: **Li + F**

The electrostatic attraction between cations and anions results in an organized array of ions in an **ionic crystal.**

The cations and anions are held by attractive electrostatic forces (given by Coulomb's law).



A Ionic bonding

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$$\text{Electrostatic Energy} \propto \frac{\text{charge A} \times \text{charge B}}{\text{distance}}$$

Electrostatic Energy is a type of potential energy, since it is based on the position of the charged particles relative to each other. If the two particles have opposite charges (as in ionic compounds), the value of electrostatic energy will be **negative**.

Energy will be ***released*** by bringing the particles together.

The **lattice energy** is the energy released when the gaseous ions combine to form the ionic solid, and is proportional to the electrostatic energy.

$$\text{Electrostatic Energy} \propto \frac{\text{charge A} \times \text{charge B}}{\text{distance}} \quad \propto \textbf{Lattice Energy}$$

$Na^+(g) + Cl^-(g) \rightarrow Na^+Cl^-(s)$ $\quad \Delta H = -770$ kJ/mol

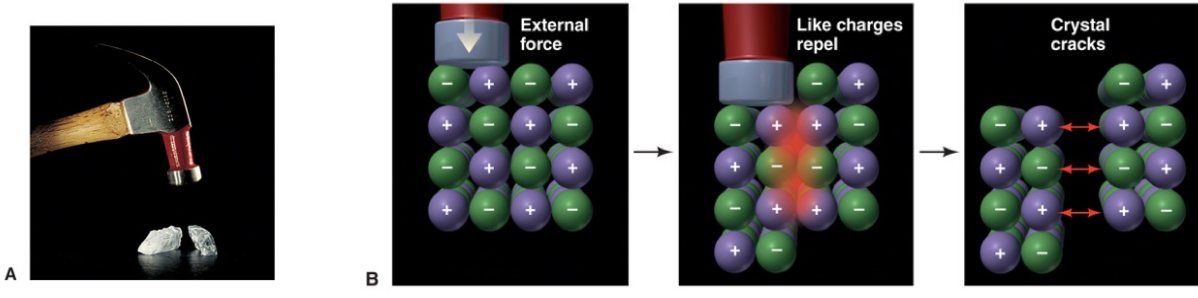
Example: Explain the trend in mp for NaCl (801 °C), NaI (661°C), and MgO (2852 °C).

Explain how this model accounts for the physical properties of ionic solids:

- hard and rigid
- brittle

Figure 8.10 **Why ionic compounds crack.**

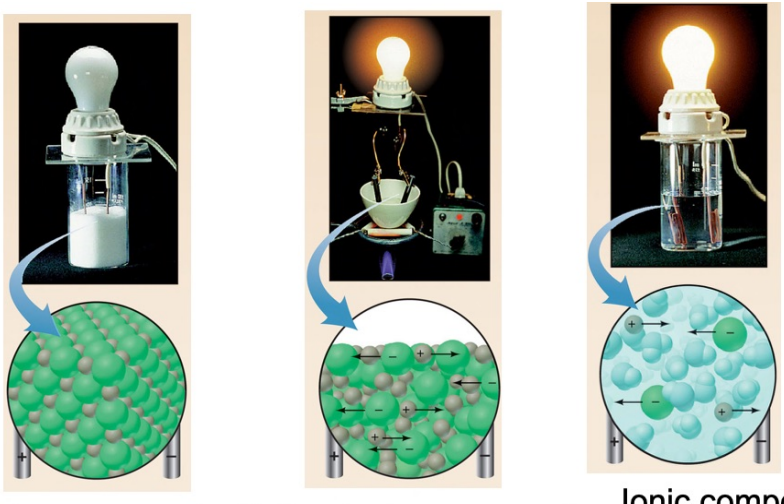
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- poor electrical conductors in the solid, but good conductors when molten or dissolved in solution

Figure 8.11

Electrical conductance and ion mobility.



Solid ionic compound | Molten ionic compound | Ionic compound dissolved in water

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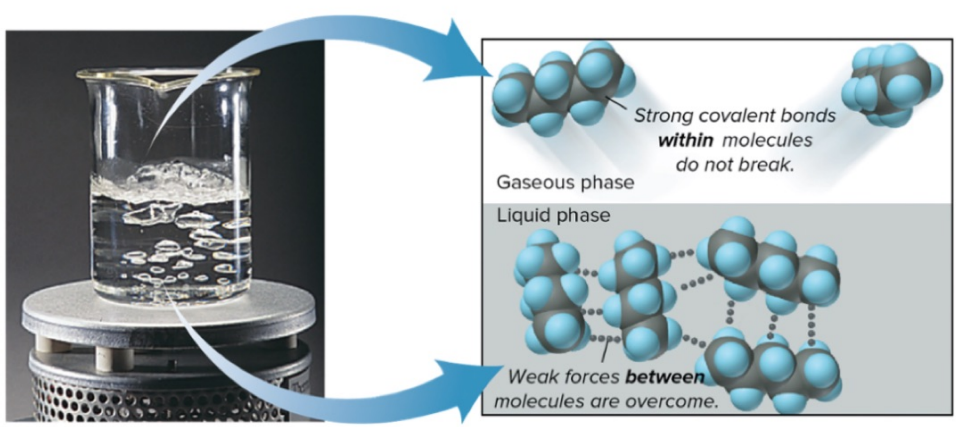
Ionic bonding is generally strong (high melting points and boiling points).

(2) Covalent bonding

- nonmetals + nonmetals
- electrons are shared between atoms (localized in the space between them)
- bonded atoms form a molecule

Consider the formation of HF(g). Use Lewis symbols to portray valence electrons and ensure that the octet rule is followed.

Covalent compounds typically exist as liquids or gases with low mp or bp. This is harder to explain, because we need to consider the distinction between the strong covalent bonds within molecules and the weak forces between molecules.



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Figure 8.17

Strong forces within molecules and weak forces between them.

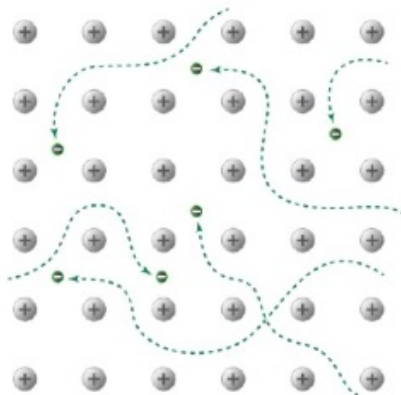
3. Metallic Bonding

Electron-sea model: all metal atoms contribute their valence electrons to form a delocalized electron "sea" throughout the metal.

·metals + metals

·electrons are highly delocalized and shared by all atoms

·bonded atoms are held together by a "sea of electrons" in an extended structure



gross simplification

Explain how this model accounts for the physical properties of metals.

- malleable and ductile,
- have moderately high mp and bp, and conduct heat and electricity well.

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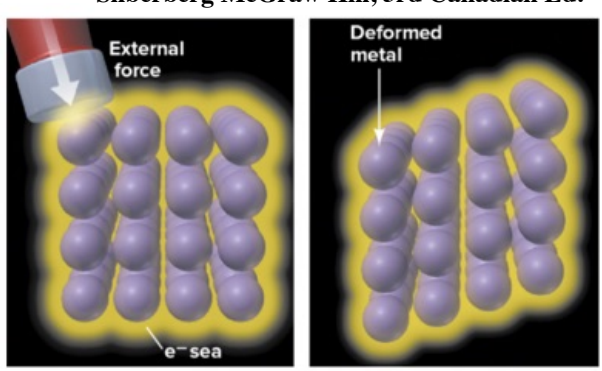


Figure 8.33 **Why metals dent and bend rather than crack.**

B. Covalent Bonding (Sec. 7.2)

Based on attractive/repulsive forces:

- Nuclear-nuclear repulsion
- Electron-electron repulsion
- Nuclear – electron attraction

Covalent bond formation in H_2.

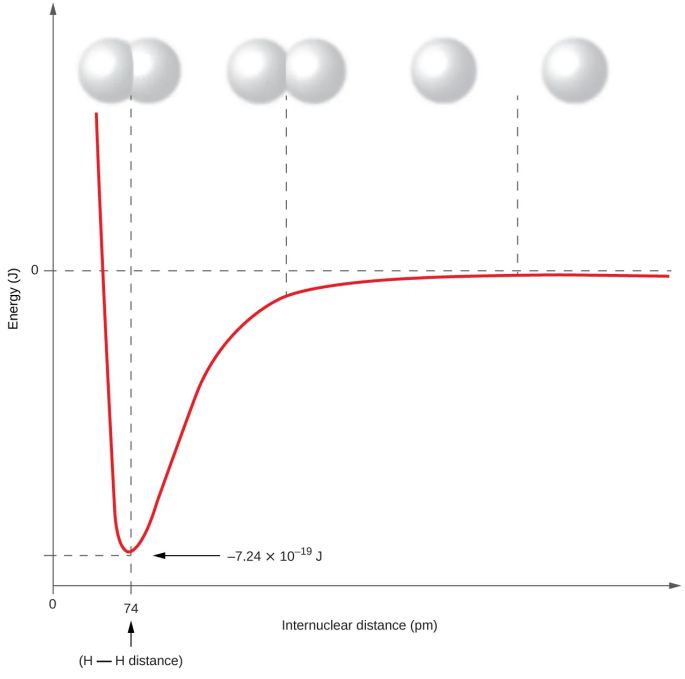


Figure 7.4 The potential energy of two separate hydrogen atoms (right) decreases as they approach each other, and the single electrons on each atom are shared to form a covalent bond. The bond length is the internuclear distance at which the lowest potential energy is achieved.

(1) No interaction
(2) Nucleus-electron attraction
(3) Optimal distance—minimal energy = bond length
(4) electron-electron and nucleus-nucleus repulsion

$$2\mathrm{H}(g) \longrightarrow \mathrm{H_2}(g) \qquad \Delta H = -436\ \mathrm{kJ}$$

$$\mathrm{H_2}(g) \longrightarrow 2\mathrm{H}(g) \qquad \Delta H = 436\ \mathrm{kJ}$$

1. Bond Energy (Section 7.5)

Bond Energy (BE or D) = standard enthalpy change for breaking a bond in one mole of gaseous molecules.

$$XY\,(g) \longrightarrow X\,(g) + Y\,(g) \qquad D_{X-Y} = \Delta H°$$

Bond energy is always positive. (Energy put into system to break bond).

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Table 8.2 Average Bond Energies (kJ/mol) and Bond Lengths (pm)

Bond	Energy	Length	Bond	Energy	Length	Bond	Energy	Length	Bond	Energy	Length
Single Bonds											
H—H	432	74	N—H	391	101	Si—H	323	148	S—H	347	134
H—F	565	92	N—N	160	146	Si—Si	226	234	S—S	266	204
H—Cl	427	127	N—P	209	177	Si—O	368	161	S—F	327	158
H—Br	363	141	N—O	201	144	Si—S	226	210	S—Cl	271	201
H—I	295	161	N—F	272	139	Si—F	565	156	S—Br	218	225
			N—Cl	200	191	Si—Cl	381	204	S—I	~170	234
C—H	413	109	N—Br	243	214	Si—Br	310	216			
C—C	347	154	N—I	159	222	Si—I	234	240	F—F	159	143
C—Si	301	186							F—Cl	193	166
C—N	305	147	O—H	467	96	P—H	320	142	F—Br	212	178
C—O	358	143	O—P	351	160	P—Si	213	227	F—I	263	187
C—P	264	187	O—O	204	148	P—P	200	221	Cl—Cl	243	199
C—S	259	181	O—S	265	151	P—F	490	156	Cl—Br	215	214
C—F	453	133	O—F	190	142	P—Cl	331	204	Cl—I	208	243
C—Cl	339	177	O—Cl	203	164	P—Br	272	222	Br—Br	193	228
C—Br	276	194	O—Br	234	172	P—I	184	246	Br—I	175	248
C—I	216	213	O—I	234	194				I—I	151	266
Multiple Bonds											
C=C	614	134	N=N	418	122	C≡C	839	121	N≡N	945	110
C=N	615	127	N=O	607	120	C≡N	891	115	N≡O	631	106
C=O	745 (799 in CO_2)	123	O_2	498	121	C≡O	1070	113			

Bond Order, Energy, Length

Increasing bond order ⟶ increasing bond energy
more e- are shared ⇒ stronger attraction (more energy is needed to break the bond)

Increasing bond order ⟶ shorter bond length.

Single versus double versus triple bonds: these do not scale as exactly double or triple values because the bonds are not exactly the same. (we shall see later why)

Besides a single covalent bond, stronger multiple bonds can also form. The bond strength is related to its length and energy.

	ethane	ethylene	acetylene
structure			
bond			
bond order			
bond length			
bond energy			

When one atom bonds to various atoms in a group, the bond strength typically decreases as we move down the group.

Down the group --> greater bond length--> lower bond energy (In general, a longer bond is a weaker bond).

Example: Rank the relative bond lengths and strengths: Si–F, Si–C, Si–O.

In a chemical reaction, atoms are rearranged when bonds are broken in reactant molecules ( ) and made in product molecules ( ). The net energy (or enthalpy) change results from the difference in bond energies:

Use $\underline{\text{BE to calculate approximate } \Delta H^{\circ}_{rxn}}$

Recall: Gas phase only for BE calculations

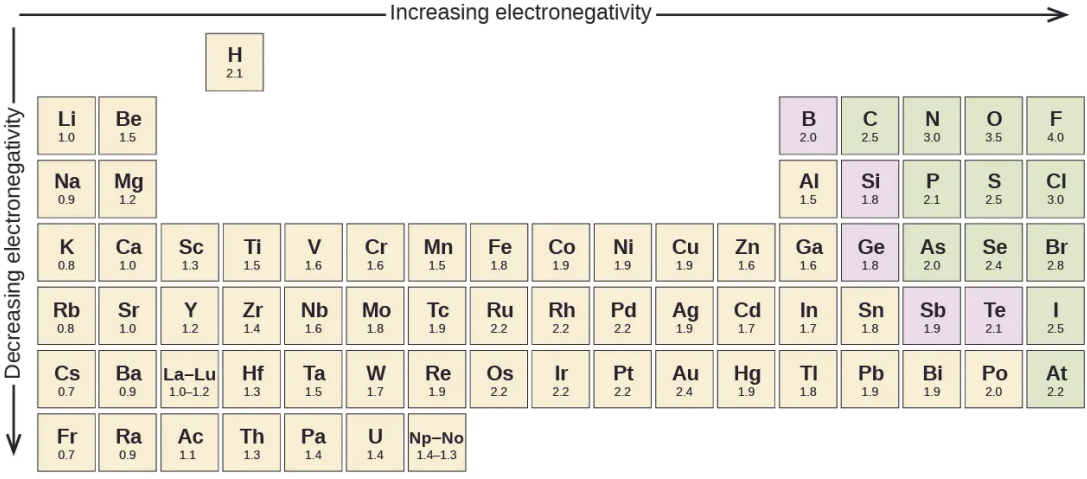
Compare the enthalpy of combustion for methane versus methanol. Which is a better fuel (a substance that reacts with oxygen to release a lot of energy)? Why?

**In general, *weaker bonds* in reactants lead to better fuels. Fuels that contain more carbon and less oxygen release more energy and are better fuels.

2. Electronegativity (Section 7.2)

EN = The relative ability of a bonded atom to attract shared electrons.

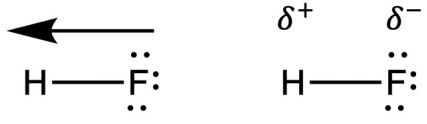
Note: be careful not to confuse **electronegativity and electron affinity**. The electron affinity of an element is a measurable physical quantity, namely, the energy change when an isolated gas-phase atom acquires an electron, measured in kJ/mol. Electronegativity describes how tightly an atom attracts electrons in a **bond**.



General trend: EN Increases

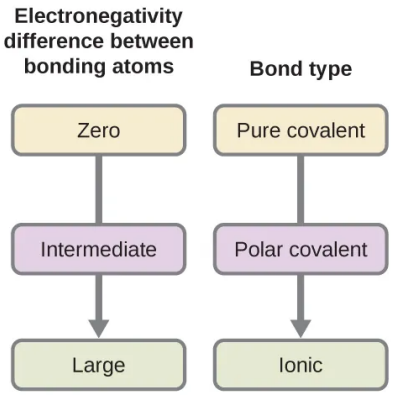
Across a period

Up a group



--H and F do not share the bond pair equally.
--F is more EN thus it gets a greater share of electrons than H.

The bond has partial ionic character and is polar covalent. The bond has a dipole moment.



Bond Type	Electronegativity Difference
pure covalent	< 0.4
polar covalent	between 0.4 and 1.8
ionic	> 1.8

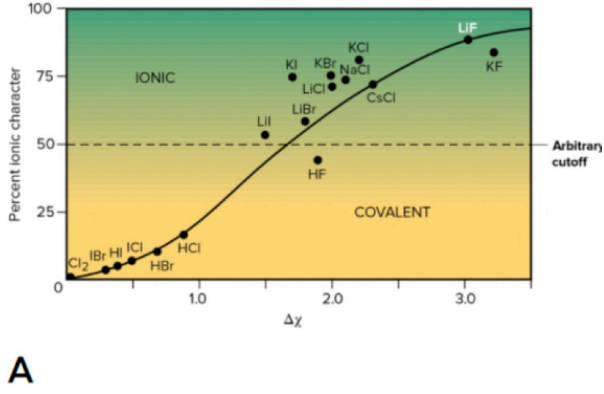


Figure 8.27 **Percent ionic character as a function of $\Delta\chi$.**
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Depending on the shape of the molecule, dipoles may or may not cancel.

Examples: CO_2 and H_2O

C. Lewis Structures (Sec. 7.3, 7.4)

1) Writing Lewis Structures

- Only **valence electrons** are shown. For main-group elements, the number of valence electrons equals the "group number" or "group number minus 10." The nucleus and core electrons are represented by the symbol of the element, and the valence electrons are represented by dots.

- A pair of **bonding electrons** (or a **bond pair**) between two atoms is a single covalent bond and is shown by a line. A pair of **non-bonding electrons** on one atom is a **lone pair** and is shown by two dots.

- Electrons are distributed so that atoms acquire a stable electron configuration, usually an **octet** (8 e^-) for most atoms (but 2 e^- for H atoms). Multiple bonds (double or triple bonds) may need to be formed.

Strategy:

1. Calculate total number of valence electrons. (Recall for main group elements, the number of valence electrons = group number or group number – 10). Add or subtract electrons if species is an anion or cation.
2. Identify central atom and write skeleton structure using SINGLE bonds.
   Carbon is usually a central atom
   Hydrogen is always a terminal atom
   If you have to choose, put the *least electronegative* atom as the central atom.
3. Determine electrons remaining. Total valence electrons – electrons needed for single bonds.
4. With remaining electrons, complete octets on TERMINAL atoms first. If any electrons are left over, complete the octet on the central atom.
5. If the central atom does not have an octet at this point, convert lone pairs on the terminal atoms into multiple bonds.

H_2CO	# valence e- # bonding e- electrons remaining:

Skeleton Structure	Completed Lewis Structure	Formal Charges

CN^-	# valence e- # bonding e- electrons remaining:

Skeleton Structure	Completed Lewis Structure	Formal Charges

NOCl	# valence e- # bonding e- electrons remaining:

Skeleton Structure	Completed Lewis Structure	Formal Charges

Formal Charge

Calculating Formal Charge:

1. Assign electrons to each atom.

Unshared electrons (lone pairs) are assigned to the atom on which they lie.

Bonding electrons get split between the atoms.

2. The difference between the number of valence electrons in the free atom and the number of assigned electrons is the formal charge.

Formal Charge of an atom =

(# of valence e-) – (# of unshared valence e- + ½ # of shared valence e-)

3. The sum of the formal charges for a neutral molecule =0

The sum of the formal charges for an ion = the charge on the ion.

**Calculate formal charges for all atoms in each example above.

Using Formal Charge:

Use formal charge to pick the best Lewis structure and/or assess the validity of your structure.

Keep formal charge as small as possible.

Keep negative formal charge on the more electronegative atom if possible.

STUDY SHEET FROM LAST WEEK

Reaction of Group I and II metals with water:

Watch the demonstration: https://www.youtube.com/watch?v=eaChisV5uR0

→ M(s) → $H_2O_{(l)}$

For an **alkali metal** reacting with water, fill in the below:

reduction reaction: $2H_2O_{(l)} + 2e^- \rightarrow H_{2(g)} + 2OH^-_{(aq)}$

oxidation reaction: $2M_{(s)} \rightarrow 2M^+_{(aq)} + 2e^-$

full reaction: $2M_{(s)} + 2H_2O_{(l)} \rightarrow 2M^+_{(aq)} + 2OH^-_{(aq)} + H_{2(g)}$

*know

- For an **alkaline earth metal** reacting with water, fill in the below:

reduction reaction: $2H_2O_{(l)} + 2e^- \rightarrow H_{2(g)} + 2OH^-_{(aq)}$

oxidation reaction: $M_{(s)} \rightarrow M^{2+}_{(aq)} + 2e^-$

full reaction: $M_{(s)} + 2H_2O_{(l)} \rightarrow M^{2+}_{(aq)} + 2OH^-_{(aq)} + H_{2(g)}$

*know

- For both alkali and alkaline earth metals, describe how reactivity changes down the group and explain: Video shows reactivity increases down the group. This makes sense in terms of ionization energy. IE decreases down the group (easier to oxidize).
- Are these metals good oxidizing or reducing agents? Why?

Alkali & Alkaline Earth (Group 1) (Group 2) → Low IE values! → easy to oxidize → Good Reducing agents (Good RA!)

Oxidizing abilities of halogens:

- The elemental halogens occur as diatomic molecules.
- Watch the demonstration: https://www.youtube.com/watch?v=u2ogMUDBaf4

- Describe how reactivity changes down the group and explain: Halogens have large magnitude EAs (tend to gain e⁻). Reactivity decreases down the group because EA magnitude decreases.
- Are halogens good oxidizing or reducing agents? Why?

Halogens gain e⁻ (reduce in reactions) due to larger magnitude EA → Good oxidizing agents